Chapter 3 : Producing Data

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⇒ Statistics is concerned with producing data as well as interpreting already available data.

→ A retail chain considering locations for new stores looks carefully at the data of local population : age, income, ethnic mix, and other variable help managers choose location where stores will prosper. This is exploratory data analysis.

⇒ If we carefully design the production of data on a smaller scale, we can use such data to draw conclusions about a wider setting.

⇒ Statistical inference uses data to answer specific questions about a setting that goes beyond the data in hand, and it attaches a known degree of confidence to its answer.

○ Observation and experiment :-

⇒ We want to know "What percent of American adult agree that the economy is getting better?"

→ We can't afford to ask all adults, so we put the question to a sample chosen to represent entire adult population.

⇒ Our goal in choosing a sample is picture of the population, that is disturbed as little as possible by the act of gathering information.

⇒ Sample surveys are one kind of observational study.

→ Other kinds of observational studies might watch the behaviour of consumers looking at the store displays, or interaction between managers and employee.

⇒ In other study we might just gather data from an experiment.

→ Here we don't just observe individuals or ask them questions.

→ We actively impose some treatment to observe the response.

→ ex: "Which TV ad will sell more toothpaste?", we show each ad to a seperate group of customers and note whether they buy the toothpaste.

○ Observation vs Experiment :

⇒ An observational study observes individuals and measures variables of interest, but does not attempt to influence the responses.

⇒ An experiment deliberately imposes some treatment on individuals to observe

their response.

⇒ When our goal is to understand cause and effect, experiments are the only source of convincing data.

☐ Example 3.1: Do cellphones cause brain cancer :-

Q) Are the observed cases of cell phone users with brain cancer evidence that cell phones cause brain cancer?

→ Careful observational studies look at large groups, not at isolated cases. To date, they find no consistent relationship between cell phone use and cancer.

⇒ But observation cannot answer cause & effect questions directly.

⇒ To see if cell phones do cause brain cancer, we might carry an experiment.

- → Choose two similar groups of people.
- → Expose one group to cell phone radiation regularly & forbid the other group any close contact with cell phones.
- → Wait a few years & then compare the rate of brain cancer in the two groups.
- → This experiment would answer our question, but is neither practical nor ethical
  - → So experiments are performed on rats.

When we simply observe cell phone use & brain cancer, any effect of radiation on the occurence of brain cancer is confounded (mixed up) with such lurking variables as age, occupation & place of residence.

Confounding : Two variables (explanatory variables or lurking variables) are confounded when their effects on a response variable cannot be distinguished from each other.

o Designing Samples :

⇒ We gather information only on a part of group to draw conclusions about the whole.

° Population → Sample :

⇒ The population in a statistical study is the entire group of individuals about which we want information.

⇒ A sample is a part of the population from which we actually

collect information, used to draw conclusions about the whole.

⇒ The design of the sample refers to the method used to choose the sample from the population.

↳ Poor sample designs can produce misleading conclusions.

EXAMPLE 3.2 **A "good" sample isn't a good idea**

A mill produces large coils of thin steel for use in manufacturing home appliances. A quality engineer wants to submit a sample of 5-centimeter squares to detailed laboratory examination. She asks a technician to cut a sample of 10 such squares. Wanting to provide "good" pieces of steel, the technician carefully avoids the visible defects in the steel when cutting the sample. The laboratory results are wonderful, but the customers complain about the material they are receiving.

EXAMPLE 3.3 **Call-in opinion polls**

Television news programs like to conduct call-in polls of public opinion. The program announces a question and asks viewers to call one telephone number to respond "Yes" and another for "No." Telephone companies charge for these calls. The ABC network program *Nightline* once asked whether the United Nations should continue to have its headquarters in the United States. More than 186,000 callers responded, and 67% said "No."

People who spend the time and money to respond to call-in polls are not representative of the entire adult population. In fact, they tend to be the same people who call radio talk shows. People who feel strongly, especially those with strong negative opinions, are more likely to call. It is not surprising that a properly designed sample showed that 72% of adults want the UN to stay.[3]

Call-in opinion polls are an example of voluntary response sampling. A voluntary response sample can easily produce 67% "No" when the truth about the population is close to 72% yes.

↳ Because people who feel strongly, especially those with strong negative opinions are more likely to put the effort to call.

- Voluntary Response Sample :-

⇒ A voluntary response sample consist of people who choose themselves by responding to a general appeal.

↳ Voluntary response samples are biased, because people with strong opinions, especially negative opions are most likely to respond.

↳ It's one common type of a bad sample design.

- Convenience Sampling: (another type of bad sample design)

⇒ It chooses individuals which are easiest to reach.

EXAMPLE 3.4 **Interviewing at the mall**

Manufacturers and advertising agencies often use interviews at shopping malls to gather information about the habits of consumers and the effectiveness of ads. A sample of mall shoppers is fast and cheap. "Mall interviewing is being propelled primarily as a budget issue," one expert told the *New York Times*. But people contacted at shopping malls are not representative of the entire U.S. population. They are richer, for example, and more likely to be teenagers or retired. Moreover, mall interviewers tend to select neat, safe-looking individuals from the stream of customers. Decisions based on mall interviews may not reflect the preferences of all consumers.[4]

↳ two biases

↳ Mall going people tend to be more towards teenage & retired side

↳ Interviewer tends to pick nicely dressed people.

⇒ Both voluntary response & convenience sample produce samples that are almost guaranteed not to represent the entire population.

- BIAS :-

⇒ The design of a study is biased if it systematically favours certain outcomes.

- Simple random sample :- (SRS)

⇒ In a voluntary response sample, people choose whether to respond. In a convinience sample, the interviewer makes the choice whom to sample.
  → In both cases personal choice produces bias.

⇒ A sample chosen by chance allows neither favoritism by the samplers nor self-selection by the respondents.
  → Choosing a sample by chance attacks all bias. black & white, rich & poor, young & old, black & white etc, all have same chance to be in the sample.

A simple random sample (SRS) of size n consists of n individuals from the population chosen in such a way that every set of n individuals has an equal chance to be the sample actually selected.

- Stratified samples :-

⇒ The general framework for statistical sampling is a probability sample.

- Probability sample is a sample chosen by chance. We must know what samples are possible & what chance/probability each possible sample has.

- Stratified Random Sample.

⇒ To select a stratified random sample, first divide the population into groups of similar individuals, called strata.
  → Then choose a seperate SRS in each stratum, and combine these SRS to form the full sample.

Choose the strata based on facts known before the sample is taken.
  → For example, a population of consumers might be divided into strata according to age, gender, & other demographic information.
    → Where there is significant overlap among strata's. A stratified design can produce more exact information than SRS. (less random)

↳ Hence avoid such cases.

- Multi-stage samples:

⇒ Another common type of probability sample chooses the sample in stages.
↳ This is usual practice for national sample of households or people.

⇒ The Current Population Survey uses a multi-stage sample design, because its impractical to to maintain all U.S households from which to select an SRS.

Stage 1: Divide the U.S into 2007 geographical areas called Primary Sampling Units, or PSUs.
↳ Select a sample of 754 PSU's. This sample includes 428 PSUs with largest population, and a stratified sample of 326 of the others.

Stage 2: Divide each PSU selected at stage 1 into small areas called "area blocks".
↳ Stratify the blocks using ethnic, and other information and take a stratified sample of the blocks in each PSU.

Stage 3: Group the housing unit in each block into clusters of four nearby units. Interview the household in a random sample of these clusters.

- Cautions about sample surveys:

⇒ For a random selection, we need an accurate & complete list of the population, because such list is rarely available, most sample suffer from some degree of undercoverage.
↳ ex:- a sample survey of households will miss homeless people & students in dormitories, or prison inmates.
↳ ex:- an opinion poll conducted by telephone will miss the 6% of american households without residential phones.

⇒ A more serious source of bias in most sample survey is non response, which occurs when an individual refuses to cooperate, or cannot be contacted.

- Undercoverage & Non Response:

⇒ Undercoverage occurs when some groups are left out of process of choosing the sample.

⇒ Non response occurs when an individual chosen for the sample can't be contacted or refuses to cooperate.

EXAMPLE 3.7 How bad is nonresponse?

The Current Population Survey has the lowest nonresponse rate of any poll we know: only about 6% or 7% of the households in the CPS sample don't respond. People are more likely to respond to a government survey such as the CPS, and the CPS contacts its sample in person before doing later interviews by phone.

The General Social Survey, conducted by the University of Chicago's National Opinion Research Center, is the nation's most important social science survey. The GSS also contacts its sample in person, and it is run by a university. Despite these advantages, its most recent survey had a 30% rate of nonresponse.

What about polls done by the media and by market research and opinion-polling firms? We don't know their rates of nonresponse, because they don't say. That itself is a bad sign. The Pew Research Center imitated a careful telephone survey and published the results: out of 2879 households called, 1658 were never at home, refused, or would not finish the interview. That's a nonresponse rate of 58%.[8]

⇒ In addition, the behaviour of the respondent or of the interviewer can cause response bias

- Respondant may lie if asked about illegal, or taboo behaviour.
- The race & sex for example for an interviewer can influence response to questions about race relations or about feminism.
- Answers to question that asks the responders to recall past events are often inaccurate because of faulty memory.

⇒ Careful training of interviewer & careful supervision to avoid variation among interviewer can greatly reduce response bias.

⇒ The wording of questions is the most important influence on the answers given to a sample survey.

- Confusing, or misleading question can introduce strong bias.

⇒ Never trust the results of a sample survey until you've read the exact questions posed.

- The amount of Non-response & the date of survey are also important.

EXAMPLE 3.8 **Should we ban disposable diapers?**

A survey paid for by makers of disposable diapers fou
opposed banning disposable diapers. Here is the actua

It is estimated that disposable diapers account for less
today's landfills. In contrast, beverage containers, thir
are estimated to account for about 21% of the trash i
opinion, would it be fair to ban disposable diapers?[10]

This question gives information on only one side of an
That's a sure way to bias the responses. A different qu
long disposable diapers take to decay and how many t
landfills each year would draw a quite different respor

- Designing Experiments :-

Subjects, Factors, and Treatment :-

⇒ The individuals studied in the experiment are often called subjects, especially if they are people.

⇒ The explanatory variables are often called factors.

⇒ A treatment is any specific experimental condition applied to the subjects.

- If the experiment has several factors, a treatment is a combination of a specific value (often called a level) of each factors.

EXAMPLE 3.11 **Absorption of a drug**

Researchers at a pharmaceutical company studying the absorption of a drug into the bloodstream inject the drug (the treatment) into 25 people (the subjects). The response variable is the concentration of the drug in a subject's blood, measured 30 minutes after the injection. This experiment has a single factor with only one level. If three different doses of the drug are injected, there is still a single factor (the dosage of the drug), now with three levels. The three levels of the single factor are the treatments that the experiment compares.



Factor A Length	Factor B Repetitions: 1 time	3 times	5 times
30 seconds	1	2	3
90 seconds	4	5	6

FIGURE 3.3 The treatments in the experimental design of Example 3.12. Combinations of levels of the two factors form six treatments.

EXAMPLE 3.12 **Effects of TV advertising**

What are the effects of repeated exposure to an advertising message? The answer may depend both on the length of the ad and on how often it is repeated. An experiment investigates this question using undergraduate students as subjects. All subjects view a 40-minute television program that includes ads for a digital camera. Some subjects see a 30-second commercial; others, a 90-second version. The same commercial is repeated either 1, 3, or 5 times during the program. After viewing, all of the subjects answer questions about their recall of the ad, their attitude toward the camera, and their intention to purchase it. These are the response variables.[15]

This experiment has two factors: length of the commercial, with 2 levels; and repetitions, with 3 levels. The 6 combinations of one level of each factor form 6 treatments. Figure 3.3 shows the layout of the treatments.

⇒ Another advantage of experiments is that we can study the combined effects of several factors simultaneously.

↳ The interaction of several factors can produce effects that maybe couldn't had been predicted looking at the effect of each factor alone.

- Comparative experiments :-

⇒ Experiments in the laboratory often have a simple design : impose the treatment & see what happens.

Subjects ⟶ Treatments ⟶ Response.

⇒ When our subjects are people & animals, confounding can happen even in controlled environment of a laboratory or medical clinic.

EXAMPLE 3.13 **Gastric freezing to treat ulcers**

"Gastric freezing" is a clever treatment for ulcers. The patient swallows a deflated balloon with tubes attached, then a refrigerated liquid is pumped through the balloon for an hour. The idea is that cooling the stomach will reduce its production of acid and so relieve ulcers. An experiment reported in the *Journal of the American Medical Association* showed that gastric freezing did reduce acid production and relieve ulcer pain. The treatment was widely used for several years. The design of the experiment was

Subjects ⟶ Gastric freezing ⟶ Observe pain relief

placebo effect

This experiment is poorly designed. The patients' response may be due to the **placebo effect**. A placebo is a dummy treatment. Many patients respond favorably to *any* treatment, even a placebo, presumably because of trust in the doctor and expectations of a cure. This response to a dummy treatment is the placebo effect.

A later experiment divided ulcer patients into two groups. One group was treated by gastric freezing as before. The other group received a placebo treatment in which the liquid in the balloon was at body temperature rather than freezing. The results: 34% of the 82 patients in the treatment group improved, but so did 38% of the 78 patients in the placebo group. This and other properly designed experiments showed that gastric freezing was no better than a placebo, and its use was abandoned.[16]

⇒ The above example could have been avoided. Experiment should compare treatments rather than attempt to asses a single treatment in isolation.

⇒ The group of patients who receive a sham treatment is called a control group, because it enables us to control the effects of lurking variable on the outcome.

- Randomized comparative experiment :

⇒ If one corn is planted on more fertile ground, or if one more drug is given to less seriously ill patients, comparisons among treatment is biased.

↳ How can we assign individuals to treatment in a way that is fair to all treatments?

⇒ The use of chance to divide the subjects in groups is also called randomization.

↳ Groups formed by randomization doesn't depend on characteristics of the subject, or on the judgment of the experimenter.

⇒ An experiment that uses both comparision & randomization is called randomized comparative experiment.

EXAMPLE 3.14 **Testing a breakfast food**

A food company assesses the nutritional quality of a new "instant breakfast" product by feeding it to newly weaned male white rats. The response variable is a rat's weight gain over a 28-day period. A control group of rats eats a standard diet but otherwise receives exactly the same treatment as the experimental group.

This experiment has one factor (the diet) with two levels. The researchers use 30 rats for the experiment and so must divide them into two groups of 15. To do this in an unbiased fashion, put the cage numbers of the 30 rats in a hat, mix them up, and draw 15. These rats form the experimental group and the remaining 15 make up the control group. *Each group is an SRS of the available rats.* Figure 3.4 outlines the design of this experiment.

We can use software or the table of random digits to randomize. Label the rats 01 to 30. Enter Table B at (say) line 130. Run your finger along this line (and continue to lines 131 and 132 as needed) until 15 rats are chosen. They are the rats labeled

05 16 17 20 19 04 25 29 18 07 13 02 23 27 21

These rats form the experimental group; the remaining 15 are the control group.

- Completely randomized design :-

⇒ There are (will see later), statistical reasons for generally using treatment groups that are about equal in size. We call design like that (Figure 3.4) completely randomized.

↳ Must have randomization & comparision.

COMPLETELY RANDOMIZED DESIGN

In a **completely randomized** experimental design, all the subjects are allocated at random among all the treatments.



FIGURE 3.4 Outline of a randomized comparative experiment, for Example 3.14.

⇒ Completely randomized designs can compare any number of treatments.

↳ Below is an example that compares three treatments.

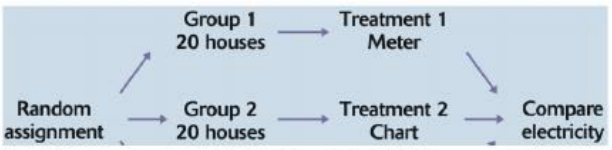




FIGURE 3.5 Outline of a completely randomized design comparing three treatments, for Example 3.15.

EXAMPLE 3.15 **Conserving energy**

Many utility companies have introduced programs to encourage energy conservation among their customers. An electric company considers placing electronic indicators in households to show what the cost would be if the electricity use at that moment continued for a month. Will indicators reduce electricity use? Would cheaper methods work almost as well? The company decides to design an experiment.

One cheaper approach is to give customers a chart and information about monitoring their electricity use. The experiment compares these two approaches (indicator, chart) and also a control. The control group of customers receives information about energy conservation but no help in monitoring electricity use. The response variable is total electricity used in a year. The company finds 60 single-family residences in the same city willing to participate, so it assigns 20 residences at random to each of the 3 treatments. Figure 3.5 outlines the design.

To carry out the random assignment, label the 60 households 01 to 60. Enter Table B (or use software) to select an SRS of 20 to receive the indicators. Continue in Table B, selecting 20 more to receive charts. The remaining 20 form the control group.

- The logic of randomized comparative experiments :-

⇒ Randomized comparative experiments are designed to give good evidence that differences in treatment actually cause the differences we see in the response.

→ Random assignment ensures assignment of subject forms groups that should be similar in all respects before treatments are applied.

→ Comparative treatment ensures that influence other than the experimental treatments operate equally on all groups.

→ Therefore, differences in average response must be due to either the treatments or to the play of chance in the random assignments of subjects to the treatments.

- That either, or deserves more thought.
  - Chance may assign the faster growing rats to one group or the other, and this creates a chance difference among groups.
    - Hence the result might depend on which group got lucky.

- Use enough subjects to reduce chance variation is the third big idea of statistical design of experiments.
  - If we assign many rats to each group, the effects of chance will average out.

PRINCIPLES OF EXPERIMENTAL DESIGN

1. **Control** the effects of lurking variables on the response, most simply by comparing two or more treatments.
2. **Randomize**—use impersonal chance to assign subjects to treatments.
3. **Replicate** each treatment on enough subjects to reduce chance variation in the results.

⇒ We want to see a difference in responses so large that it's unlikely to happen just because of chance variation

⇒ An observed effect so large that it would rarely occur by chance is called statistically significant.

⇒ A statistically significant association in data from a well-designed experiment does imply causation.

- Cautions about Experimentation :-

⇒ The study is double blind: neither the subject, nor the medical personell who worked with them knew which treatment the subject had.

⇒ The most serious weakness of an experiment is lack of realism.

→ The subjects or treatments or setting of an experiments may not realistically duplicate the conditions we want to study.

Here are two examples

EXAMPLE 3.16 **Response to advertising**

The study of television advertising in Example 3.12 showed a 40-minute videotape to students who knew an experiment was going on. We can't be sure that the results apply to everyday television viewers. The student subjects described their reactions but did not actually decide whether to buy the camera. Many experiments in marketing and decision making use as subjects students who know they are taking part in an experiment. That's not a realistic setting.

EXAMPLE 3.17 **Center brake lights**

Do those high center brake lights, required on all cars sold in the United States since 1986, really reduce rear-end collisions? Randomized comparative experiments with fleets of rental and business cars, done before the lights were required, showed that the third brake light reduced rear-end collisions by as much as 50%. Alas, requiring the third light in all cars led to only a 5% drop.

What happened? Most cars did not have the extra brake light when the experiments were carried out, so it caught the eye of following drivers. Now that almost all cars have the third light, they no longer capture attention.

- Matched Pairs design :-

⇒ Completly randomized designs are often inferior to more elaborate statistical designs.

→ In particular matching the subjects in various ways can produce more precise results than simple randomization.

⇒ One common design that combines matching with randomization is the matched pairs design.

→ It compares just two treatments.

→ Choose pairs of subject that are as closely matched as possible.

→ Assign one of the treatment to one subject randomly, and the other treatment to other subject

⇒ Sometimes each pair consist of one subject only, who gets both treatments one after the other.

↳ Each subject serves as his control.

↳ The order of treatment can influence the subjects' response, so we randomize the order of each subjects again.

EXAMPLE 3.18 **Coke versus Pepsi**

example of treatment applied one after another

Pepsi wanted to demonstrate that Coke drinkers prefer Pepsi when they taste both colas blind. The subjects, all people who said they were Coke drinkers, tasted both colas from glasses without brand markings and said which they liked better. This is a matched pairs design in which each subject compares the two colas. Because responses may depend on which cola is tasted first, the order of tasting should be chosen at random for each subject.

When more than half the Coke drinkers chose Pepsi, Coke claimed that the experiment was biased. The Pepsi glasses were marked *M* and the Coke glasses were marked *Q*. Aha, said Coke, this just shows that people like the letter *M* better than the letter *Q*. A careful experiment would in fact take care to avoid any distinction other than the actual treatments.[17]

- Block designs :-

⇒ Matched pair do not assign all the subjects at once to the two treatments.

↳ Instead only randomizes within each matched pair.

↳ This reduces the variations among subjects.

- Block design :

⇒ A block is a group of subjects that are known before the experiment to be similar in some way expected to affect the response to the treatment.

↳ In a block design, the random assignment of individuals to treatments is carried out seperately within each block.

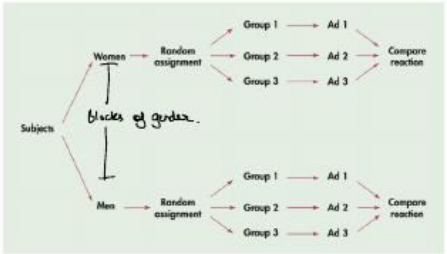
⇒ Matched pairs are an example of block design of size 1.

EXAMPLE 3.29 Men, Women, and Advertising

Women and men respond differently to advertising. An experiment to compare the effectiveness of three television commercials for the same product will want to look separately at the reactions of men and women, as well as assess the overall response to the ads.

A completely randomized design considers all subjects, both men and women, as a single pool. The randomization assigns subjects to three treatment groups without regard to their gender. This ignores the differences between men and women. A better design considers women and men separately. Randomly assign the women to three groups, one to view each commercial. Then separately assign the men at random to three groups. Figure 3.10 outlines this improved design.

FIGURE 3.10 Outline of a block design, for Example 3.29.



⇒ Block is a group of subjects formed before an experiment starts.

⇒ Block designs are similar to stratified samples.

↳ Both blocks & strata group similar individuals together.

↳ We use different names only because ideas developed separately for sampling & experimentation.

- Toward Statistical Inference :-

⇒ A poll asked a random sample of 1009 adults about eating dinner in restaurants.

↳ 60% reported that they ate dinner out at least once during the past week.

↳ What the truth about the 1009 people in sample, but what's the truth about the almost 230 million american adults.

↳ Because the samples where choosen randomly, its reasonable to think these 1009 people represents the population pretty well.

⇒ That's a basic concept in statistics : use a fact about a sample to estimate the truth about the whole population. This is called statistical inference.

↳ Because we infer conclusions about the wider population from data on selected individuals.

- Parameter & Statistics :

→ A parameter is a number that describes the population.

↳ A parameter is a fixed number, but in practice we do not know its value.

⇒ A statistic is a number that describes a sample. The value of a statistic is known when we have taken a sample, but it can change from sample to sample.

↳ We often use a statistic to estimate an unknown parameter.

CASE 3.1 **Changes in Consumer Behavior** The economic recession of 2008 led many consumers to reduce their spending on nonessential items. A Gallup Poll taken December 11–14, 2008, asked a random sample of 1008 U.S. adults, aged 18 and over, about eating out in restaurants during the previous week. The results from this survey can be compared with the results from similar surveys taken in 2007 and 2006 to assess the extent to which there has been a change.[33]

The proportion of the sample who said that they ate out in a restaurant last week is

$$\hat{p} = \frac{605}{1009} = 0.60 = 60\%$$

⇒ The number $\hat{p} = 0.60$ is a statistic. The corresponding parameter is the proportion (call it p) of all adult U.S residents who would have said that they ate out in a restaurant last week if asked the same question.

⇒ We don't know the value of p, so we use $\hat{p}$ as an estimate.

- Sampling Variability & sampling Distributions :-

⇒ If Gallup took a second random sample of 1009 adults, the new sample would have different people in it.

⇒ It's almost certain that there would not be exactly 605 positive responses.

↳ That is the value of the statistic $\hat{p}$ will vary from sample to sample.

↳ This basic fact is called sampling variability.

Q) Could it happen that one random sample finds that 60% said that they ate out in a restaurant last week, and a second random sample finds that only 40% say they ate out?

↳ If the variation is too great, the information obtained from our sample may not be very useful.

⇒ All of the statistical inference is based on one idea : to see how trustworthy a procedure is, ask what would happen if we repeated it many times.

⇒ To understand why sampling variability is not fatal, we therefore ask, "What would happen if we took many samples?" Here's how to answer that question.

↳ Assume that you have population with many individuals & take a large number of samples from this population.

↳ Calculate the sample proportion $\hat{p}$ for each sample.

↳ Make a histogram with the values of $\hat{p}$.

↳ Examine the distribution displayed in the histogram for shape, center, & spread, as well as the outliers or other deviations.

EXAMPLE 3.30 Samples of Size 100

CASE 3.1 Suppose that in fact (unknown to Gallup), exactly 60% of all adults ate out last week. That is, the truth about the population is that $p = 0.6$. What if we select an SRS of size 100 from this population and use the sample proportion $\hat{p}$ to estimate the unknown value of the population proportion p? Using software, we simulated doing this 1000 times.

Figure 3.11 illustrates the process of choosing many samples and finding $\hat{p}$ for each one. In the first sample, 56 of the 100 people say they find shopping frustrating, so $\hat{p} = 56/100 = 0.56$.

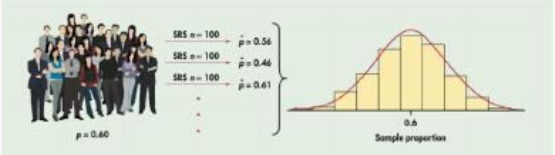


FIGURE 3.11 The results of many SRSs have a regular pattern. Here, we draw 1000 SRSs of size 100 from the same population. The population proportion is $p = 0.60$. The histogram shows the distribution of 1000 sample proportions $\hat{p}$.

Only 46 in the next sample feel this way, so for that sample $\hat{p} = 0.46$. Choose 1000 samples and

make a histogram of the 1000 values of $\hat{p}$. That's the graph at the right of Figure 3.11.

⇒ A gaussian distribution !

Q) What would happen if we used a larger sample size ?

EXAMPLE 3.31 Samples of Size 2500

CASE 3.1

Suppose that Gallup used 2500 people for their sample, not just 100. Figure 3.12 shows the process of choosing 1000 SRSs, each of size 2500, from a population in which the true sample proportion is $p = 0.6$. The 1000 values of $\hat{p}$ from these samples form the histogram at the right of the figure. Figures 3.11 and 3.12 are drawn on the same scale. Comparing them shows what happens when we increase the size of our samples from 100 to 2500. These histograms display the *sampling distribution* of the statistic $\hat{p}$ for two sample sizes.

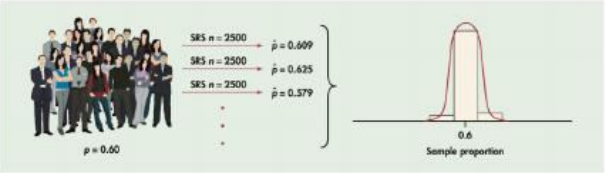


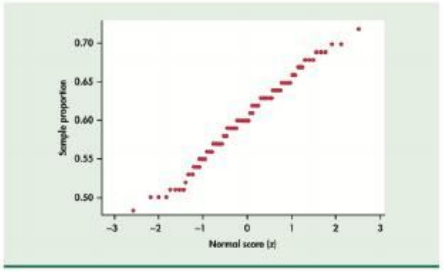
FIGURE 3.12 The distribution of the sample proportions $\hat{p}$ for 1000 SRSs of size 2500 drawn from the same population as in Figure 3.11. The two histograms have the same scale. The statistic from the larger sample is less variable.

- Sampling Distribution :-

⇒ The sampling distribution of a statistic is the distribution of values taken by the statistic in all possible samples of the same size from the same population.

✱ Shape: The histogram looks normal, however it is a Normal quantile plot of the values of $\hat{p}$ for our samples of size 100. It confirms that the distribution is close to normal. (Larger samples size is ever more close to normal).

FIGURE 3.13 Normal quantile plot of the sample proportions in Figure 3.11. The distribution is close to Normal except for some clustering due to the fact that the sample proportions from a sample of size 100 can take only values that are multiples of 0.01. Because a plot of 1000 points is hard to read, this plot presents only every 10th value.



✱ Center: In both cases the values are centered at 0.6 . (at the p of the true population)

✱ Spread: The values of $\hat{p}$ from samples of size 2500 (large sample size) are much less spread out than the values from samples of size 100. (0.0097 vs 0.051).

✱✱ Bias and Variability :-

⇒ Our simulation shows that a sample of size 2500 will almost give an estimate (with good random sample) $\hat{p}$ that is close to the truth about the population

↳ This is true for any population.

⇒ Bias concerns the center of the sampling distribution.

↳ A statistic used to estimate a parameter is the mean of its sample distribution

is equal to the true value of the parameter being estimated.
↑ Population

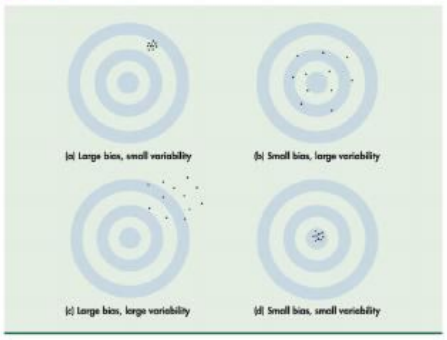
⇒ The variability of statistics is described by the spread of its sampling distribution.

→ The spread is determined by the sampling design & sample size n. Statistics from larger samples have smaller spread.

⇒ We can think of the true value of population parameter as the bull's eye on a target, and of a sample statistic as an arrow shot at the bull's-eye.

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FIGURE 3.14 Bias and variability in shooting arrows at a target. Bias means the archer systematically misses in the same direction. Variability means that the arrows are scattered.



⇒ Bias means that the aim is off, and the arrows land consistently off the bull's eye in the same direction.

⇒ Large variability means that repeated shots are widely scattered on the target.

○ Managing Bias and Variability :-

⇒ To reduce bias (bias ⇒ going all in one direction in sampling distribution), use random sampling.

→ The values of statistics computed from SRS, neither consistently overestimate, nor consistently underestimate the value of the population parameter.

⇒ To reduce the variability of statistics from an SRS, use a larger sample.

→ You can make the variability as small as you want by taking a large enough sample.

○ Population size don't matter as long as it's atleast 100 times larger than the sample :-

⇒ The variability from a random sample does not depend on the size of the population, as long as the population is atleast 100 time larger than the sample.